IIT MADRAS · SOFTWARE ENGINEERING · SEPT 2026

Milestone 2 Report

Scheduling, Design and Frontend

Student Mentoring and Career Support Platform

Problem Statement 2

Team 032 – Himalayan Coders

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1. Project Schedule and Sprint Plan

1.1 Scheduling tool

The team plans and tracks its work in **Jira** (project key KAN). Epics group the work of each milestone or sprint. Tasks carry one owner, a start date and a due date, and move across the board columns To Do, In Progress, In Review and Done. The Jira **Timeline** view is our Gantt chart and the **Board** view is our Kanban board. Each task records its owner, so both views show every member's contribution.

1.2 Team roles

Member	Roles
Himanshu Rathor	Scrum Master, Product Manager
Ankur Maurya	Backend Developer, Frontend Developer
Hemendra Kumar Sharma	Frontend Developer, Backend Developer
Adnan Mustafa	Tester, Code Reviewer
Bojannagari Pavan Kumar Reddy	Code Reviewer, Tester

1.3 Overall schedule

Phase	Dates	Focus
Milestone 1	until 25 Oct	User requirements
Milestone 2	26 Oct – 1 Nov	Scheduling, design, frontend pages
Sprint 1 (Milestone 3)	2 Nov – 22 Nov	APIs, test cases and user testing for the core flow
Sprint 2 (Milestone 4)	23 Nov – 6 Dec	APIs, test cases and user testing for the rest of the flow
Milestone 5	7 Dec – 20 Dec	Final report, demo video, working prototype

1.4 Sprint 1 (2 Nov – 22 Nov): core client-mediated flow

Jira task	User stories	Dates
Registration and login APIs	US-S01, US-M01	2 – 8 Nov
Student and mentor profile APIs	US-S02, US-M02, US-C05	6 – 12 Nov
Admin dashboard and role-based access APIs	US-C01, US-C06	9 – 15 Nov
Match creation APIs	US-C02	12 – 18 Nov
Test cases, pytest and Swagger YAML	all of the above	9 – 20 Nov
User feedback session and Sprint 2 plan	–	20 – 22 Nov

1.5 Sprint 2 (23 Nov – 6 Dec): confirmation, tracking and student views

Jira task	User stories	Dates
Match confirmation and notification APIs	US-C03	23 – 29 Nov
Student goals, preferences and availability APIs	US-S03 to US-S06	23 – 29 Nov
Match status management and overview APIs	US-C04, US-C07	26 Nov – 2 Dec
Student connection view, readiness, concern, feedback and status APIs	US-S07 to US-S11	28 Nov – 3 Dec
Test cases, pytest and Swagger YAML	all of the above	30 Nov – 5 Dec
User feedback session and final-phase plan	–	4 – 6 Dec

All 21 user stories from Milestone 1 are covered across the two sprints.

1.6 Iterations

Each sprint is split into weekly iterations. Every sprint ends with a user feedback session whose findings shape the next sprint plan. Tasks are reviewed by the Code Reviewers and tested by the Testers before they move to Done.

1.7 Gantt chart

INSERT: Jira Timeline screenshot (Weeks view, epics expanded)

1.8 Kanban board

INSERT: Jira Board screenshot (cards spread across the columns)

2. Scrum Meetings

The Scrum Master leads two scrum meetings each week. Each meeting covers what was done, what is planned next and any blockers. Minutes are kept for every meeting.

Date	Attendees	Discussed / decisions	Actions (owner, due date)

3. Design of Components

The platform follows a client-mediated model. Students and mentors each maintain their own profile, but they cannot browse each other. The client (coordinator) reviews both sides, proposes a match, and tracks it until it ends.

Component	Responsibilities
Authentication and roles	Registration, login, and three roles: student, mentor and administrator. Every action is checked against the role on the server, because hiding a link in the interface is not security.
Student module	Maintain goals, skills, preferences and availability. View the assigned mentor after the match is confirmed. Give a readiness response, feedback, and raise concerns.
Mentor module	Maintain qualifications, expertise, bio and availability slots. Accept or decline match requests, and see accepted mentees.
Administrator module	Verify mentor profiles, propose matches, change match status, review concerns, and see delayed proposals and an overview by status.
Match workflow	Controls the status of a match: proposed, confirmed, active, completed or cancelled. Only allowed transitions are accepted and each change is logged.
Notifications	Tells the people involved when a match is proposed, accepted, declined or changes status.
Data layer	A relational database holding users, profiles, availability, matches, status log, feedback, concerns and notifications.

Match status flow: proposed → confirmed (mentor accepts) → active → completed. A proposed or confirmed or active match can be cancelled. A declined proposal becomes cancelled, and the coordinator can then propose a new match.

4. Class Diagram

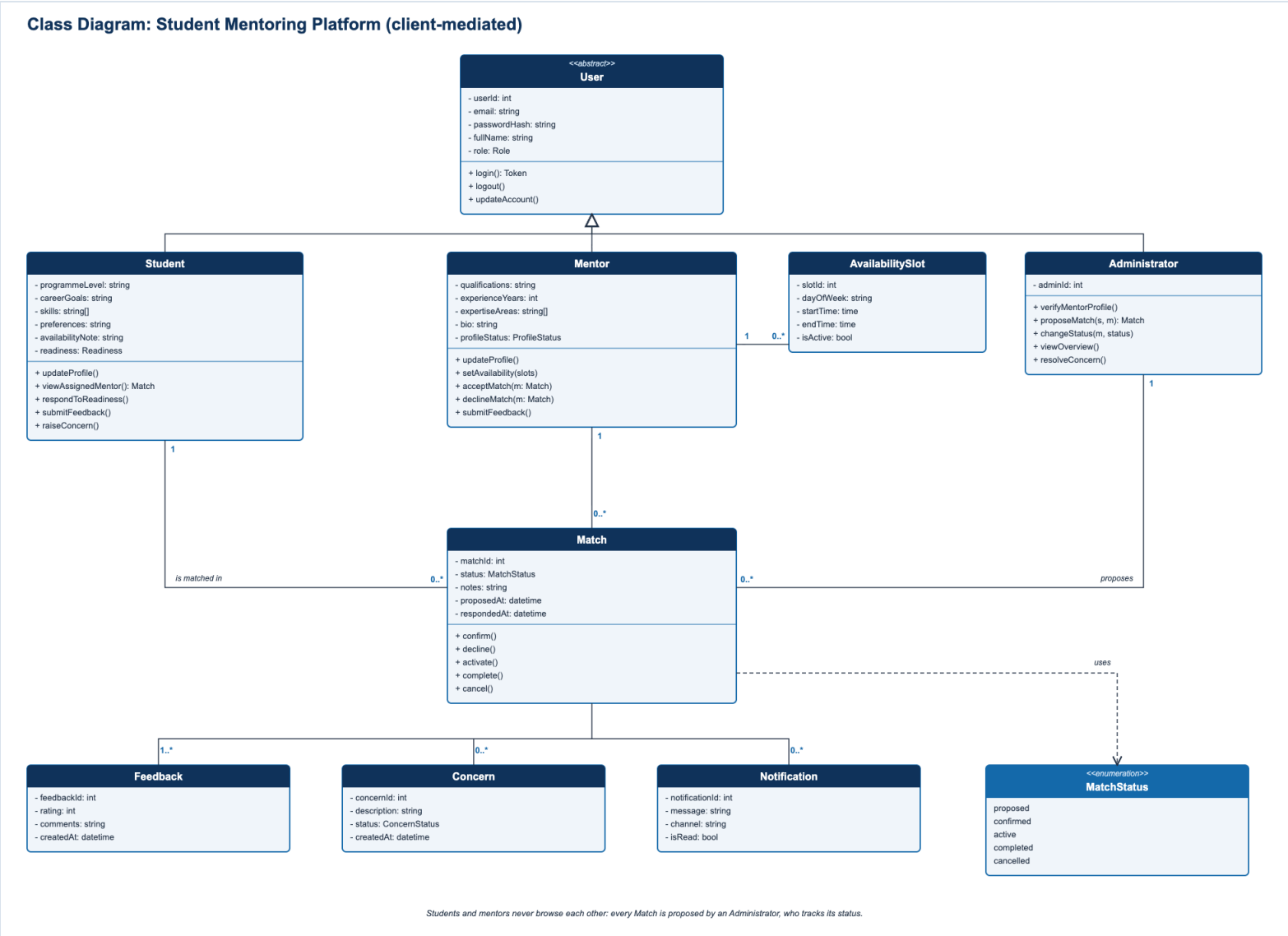


Figure 1. Class diagram. Student, Mentor and Administrator extend User. Every Match is proposed by an Administrator and links one Student and one Mentor.

5. Database Design

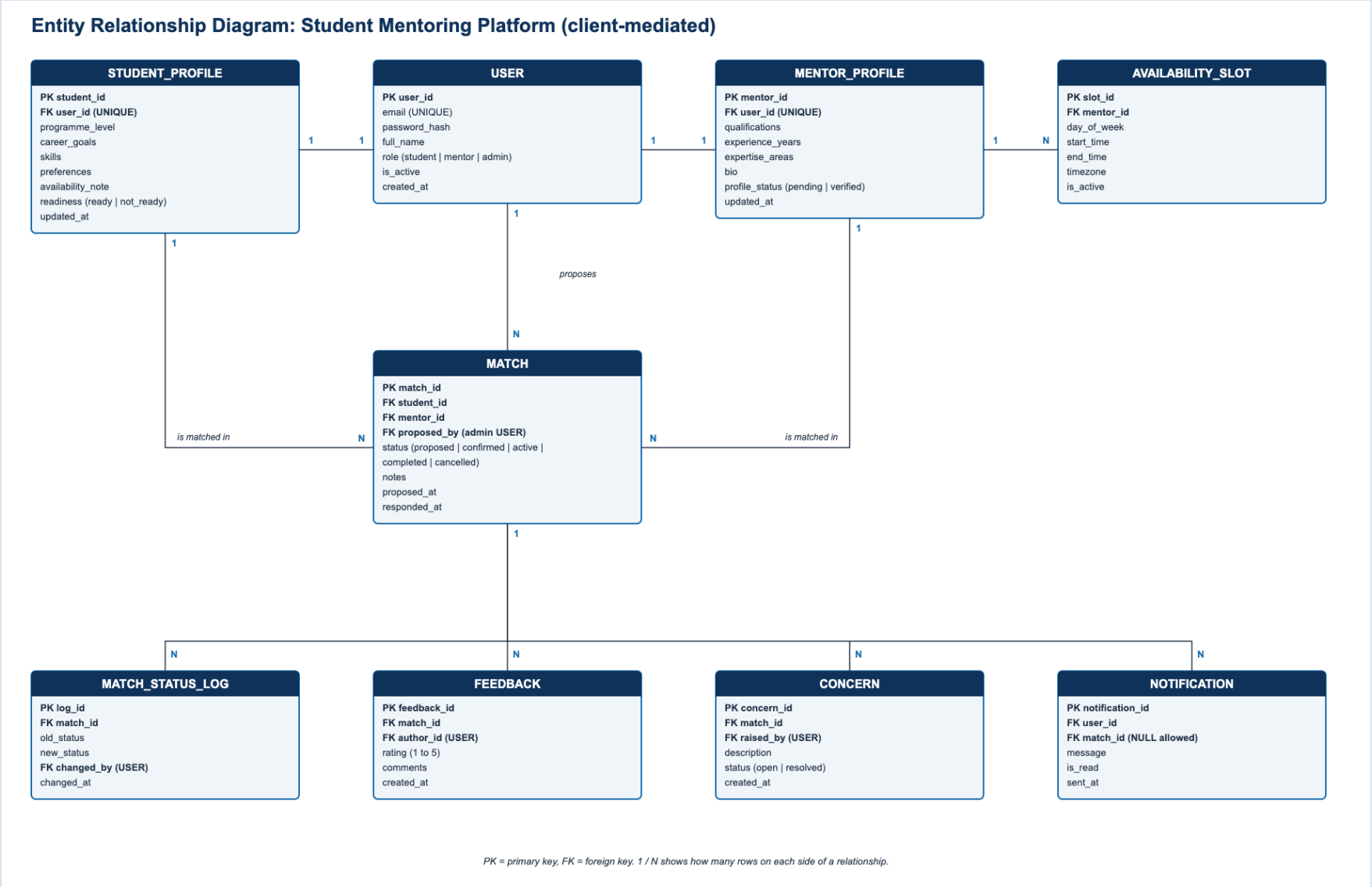


Figure 2. Entity relationship diagram.

Table	Purpose and key constraints
USER	One row per account. Email is unique. Role is student, mentor or admin.
STUDENT_PROFILE	Goals, skills, preferences and readiness. One per student user (user_id unique).
MENTOR_PROFILE	Qualifications, expertise and verification status. One per mentor user.
AVAILABILITY_SLOT	Weekly time slots of a mentor. Many slots per mentor.
MATCH	Links a student and a mentor. Records the administrator who proposed it and its status.
MATCH_STATUS_LOG	History of every status change, with who changed it and when.
FEEDBACK	Rating and comments about a match.
CONCERN	A concern raised about a match, open until the coordinator resolves it.
NOTIFICATION	Messages sent to users, optionally linked to a match.

6. Frontend Pages and Screenshots

The frontend is a set of 11 linked pages in plain HTML, CSS and JavaScript. It runs from the file `index.html` without installing anything. Sample data is kept in the browser so the whole flow can be tried: the coordinator proposes a match, the mentor accepts, and only then does the student see the mentor. There is no backend yet; the API is built in Milestones 3 and 4.

Page	Role	Purpose
index, register	all	Log in and create an account
student	student	Dashboard, match status, notifications
student-profile	student	Goals, skills, preferences, availability
student-mentor	student	Assigned mentor, readiness, feedback, concerns
mentor	mentor	Accept or decline requests, mentees
mentor-profile	mentor	Background and availability

admin	coordinator	Overview by status, delayed proposals, concerns
admin-mentors	coordinator	Verify mentor profiles
admin-matches	coordinator	Propose matches, change status

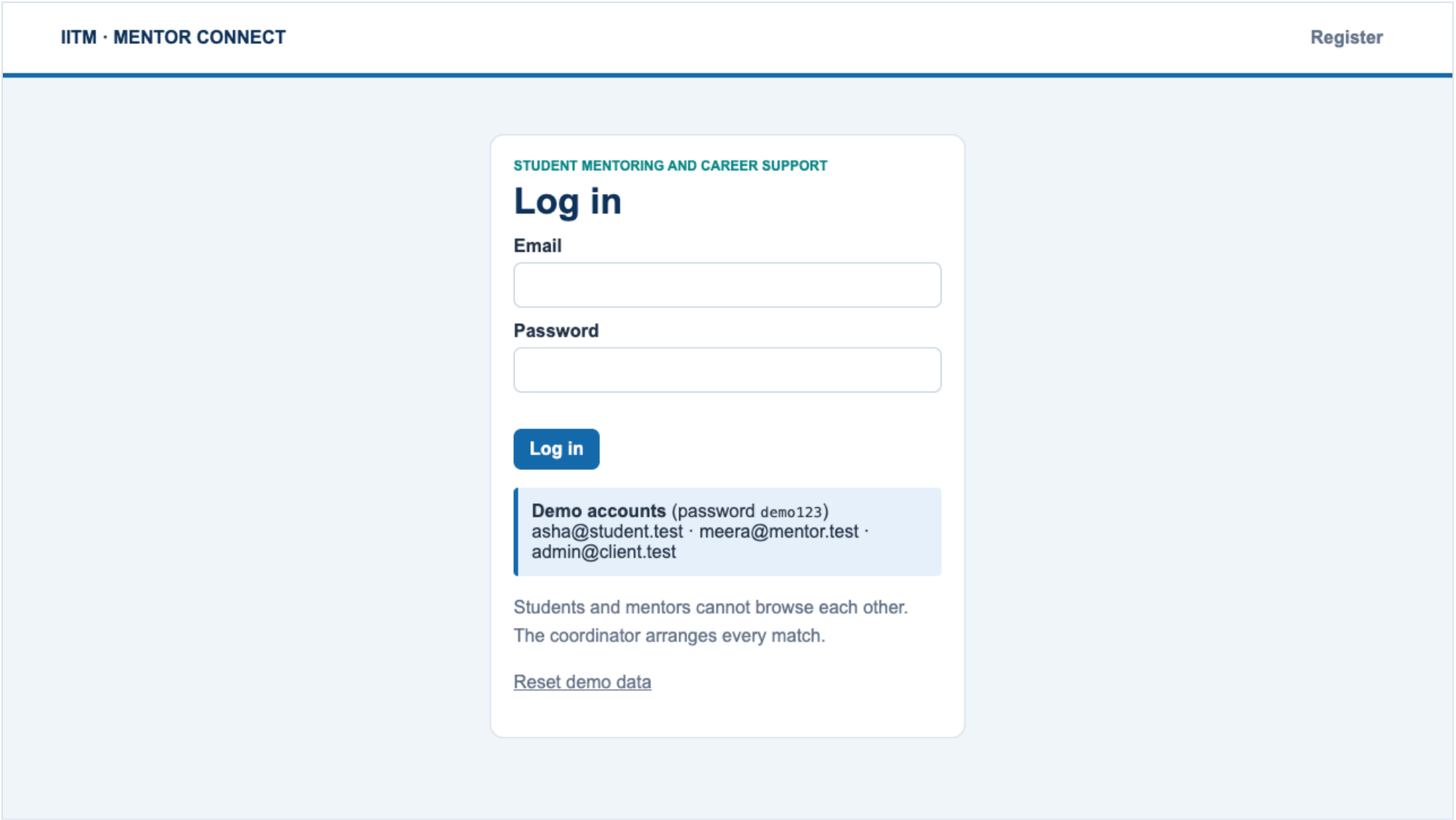


Figure 3. Login page with demo accounts.

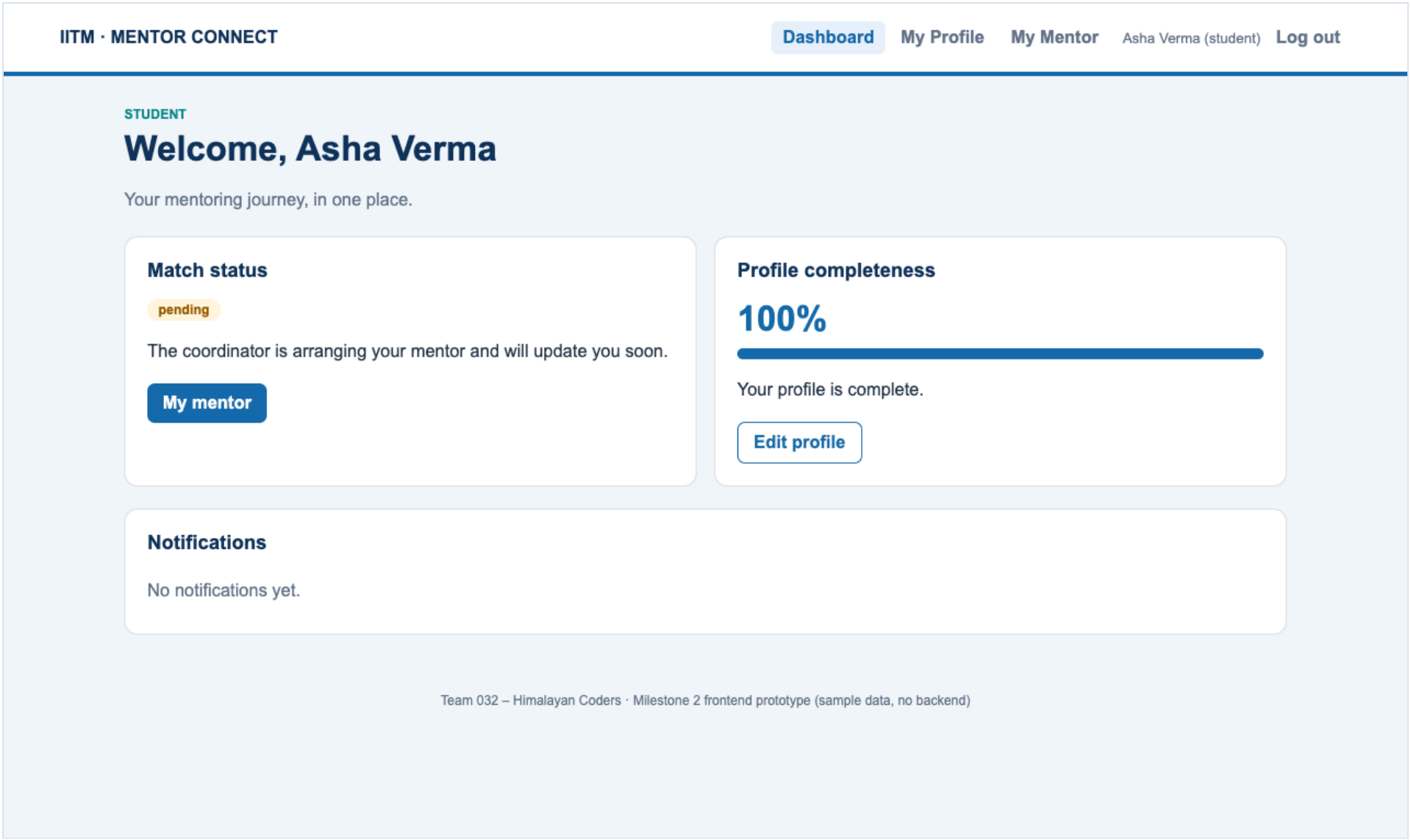


Figure 4. Student dashboard.

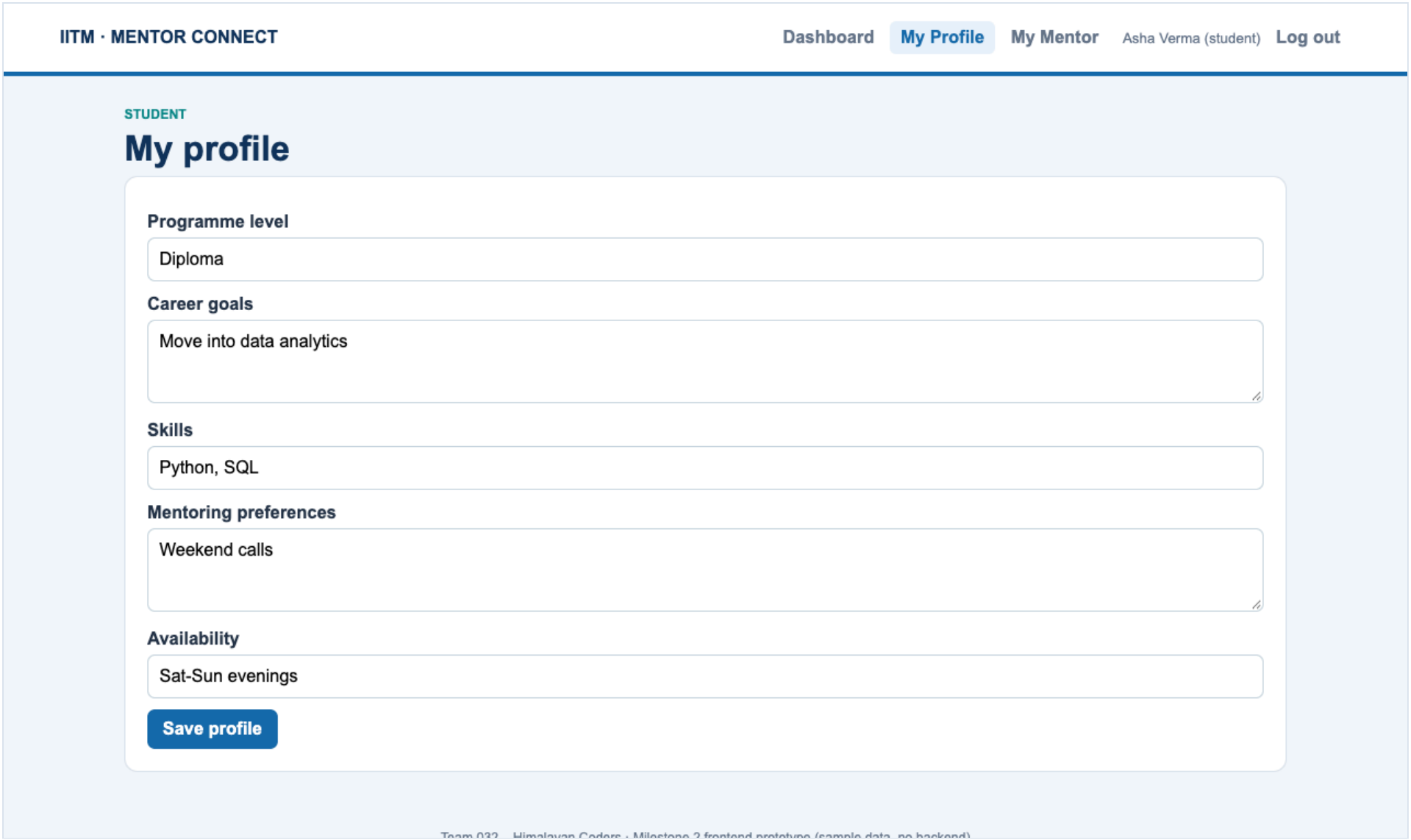


Figure 5. Student profile.

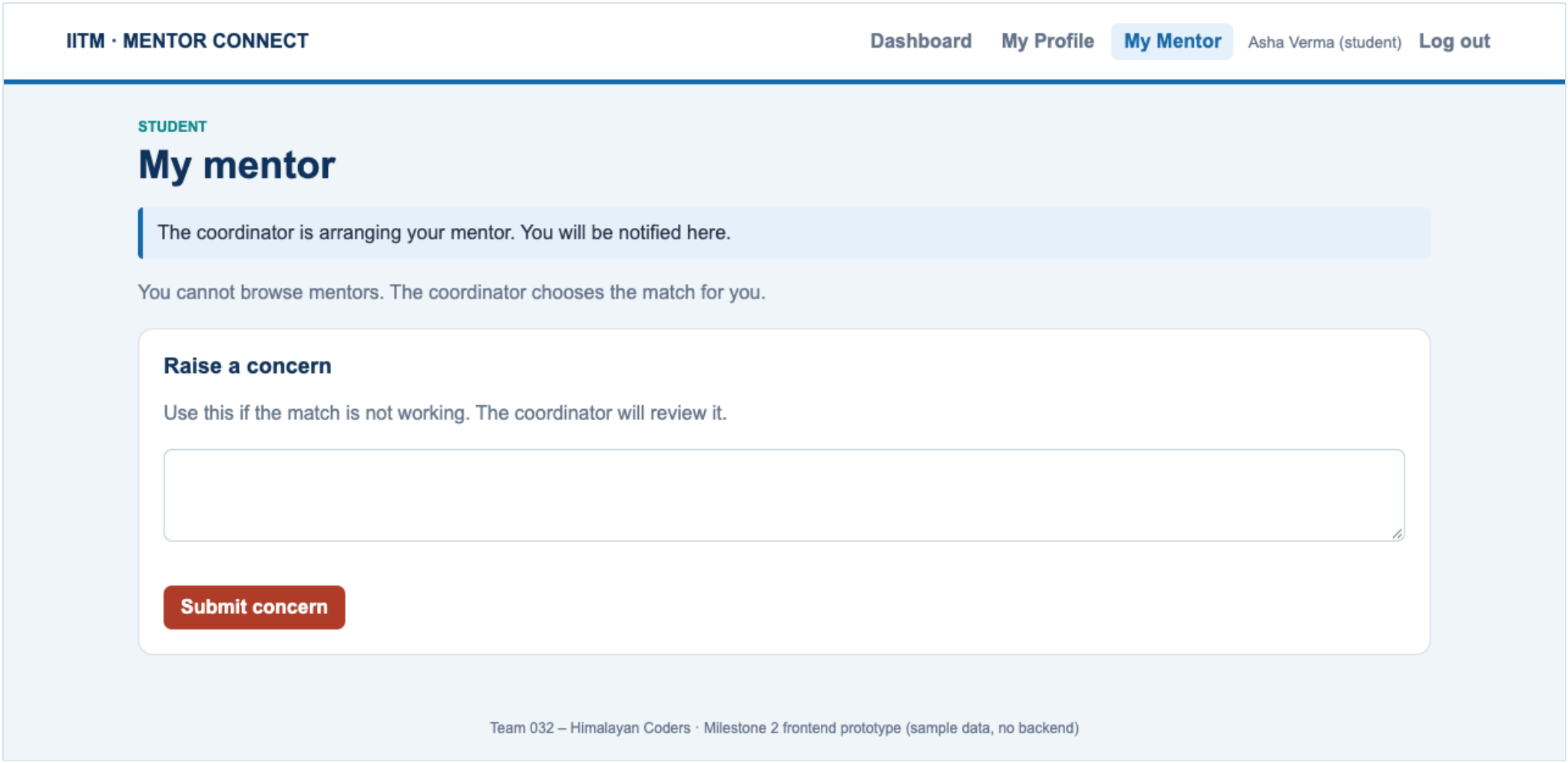


Figure 6. Student's mentor page while the coordinator is still arranging the match.

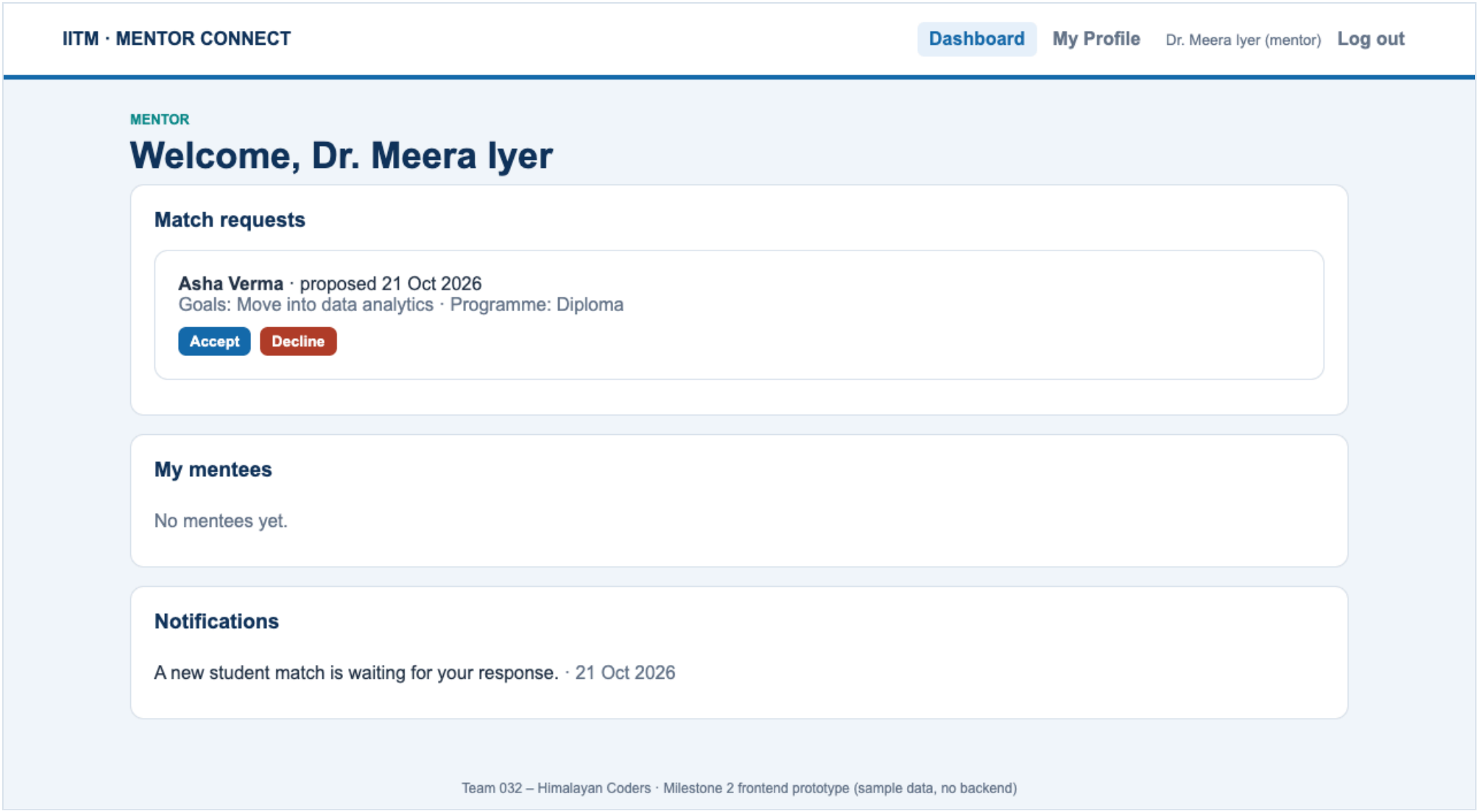


Figure 7. Mentor dashboard with a match request to accept or decline.

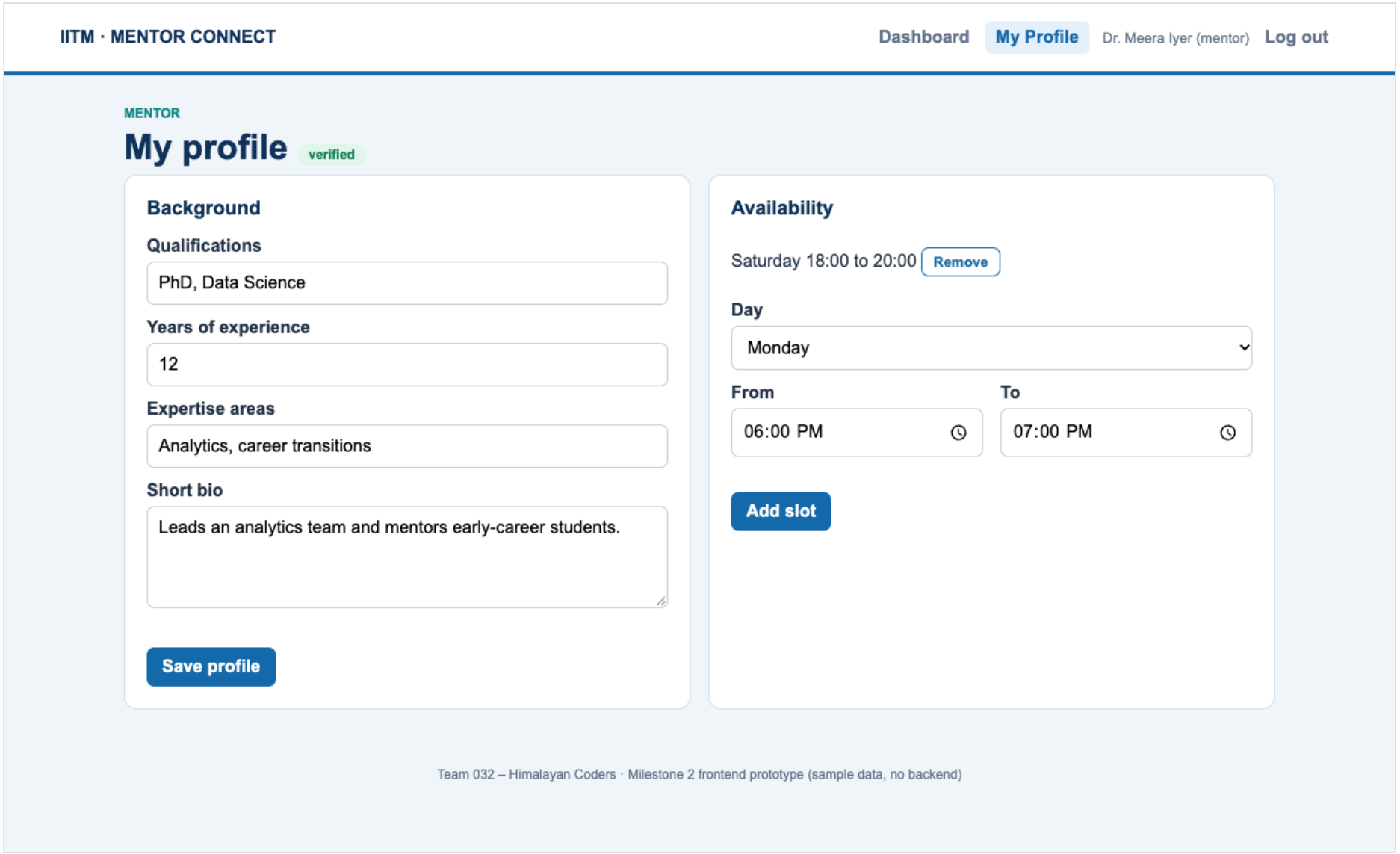


Figure 8. Mentor profile and availability.

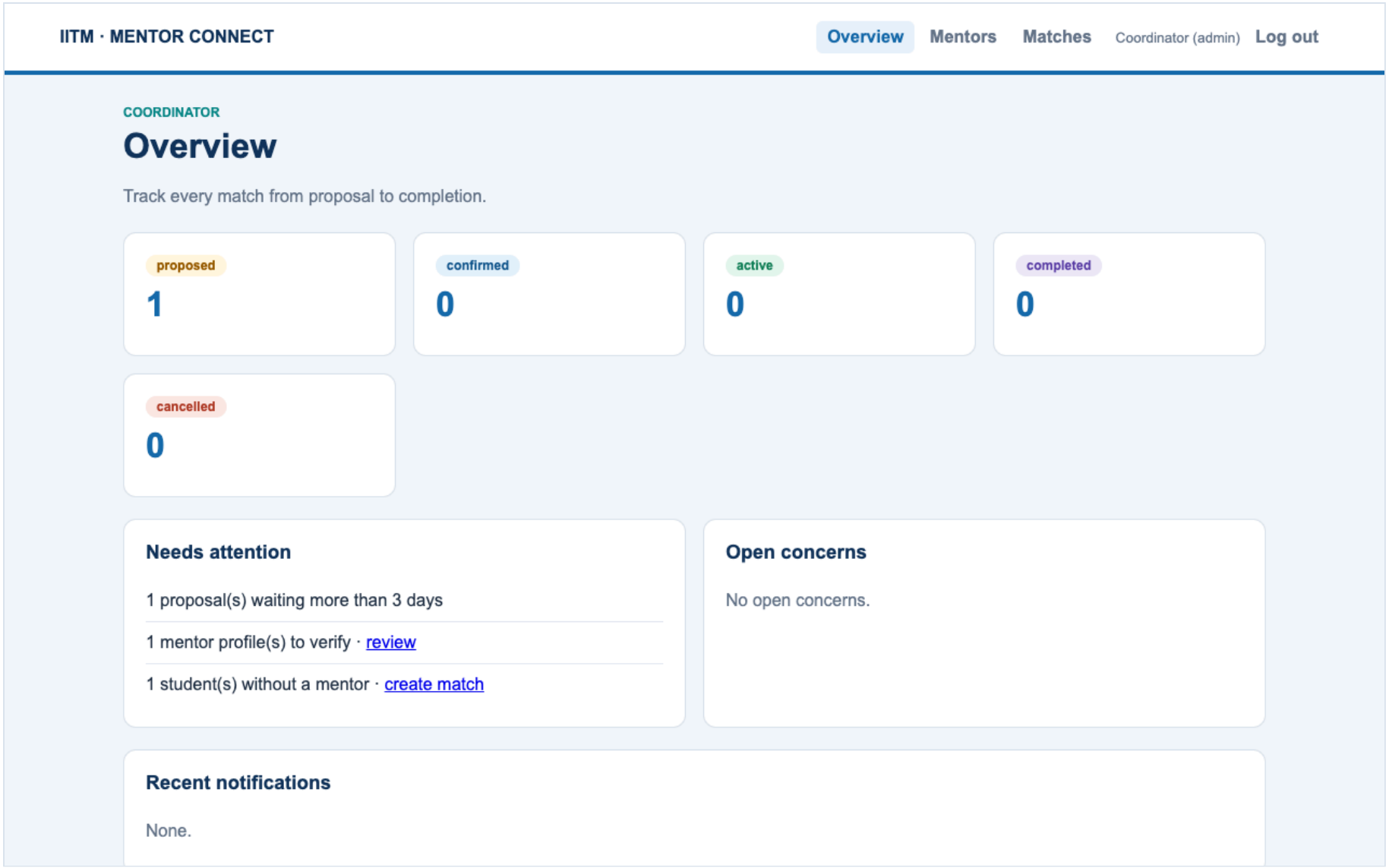


Figure 9. Coordinator overview.

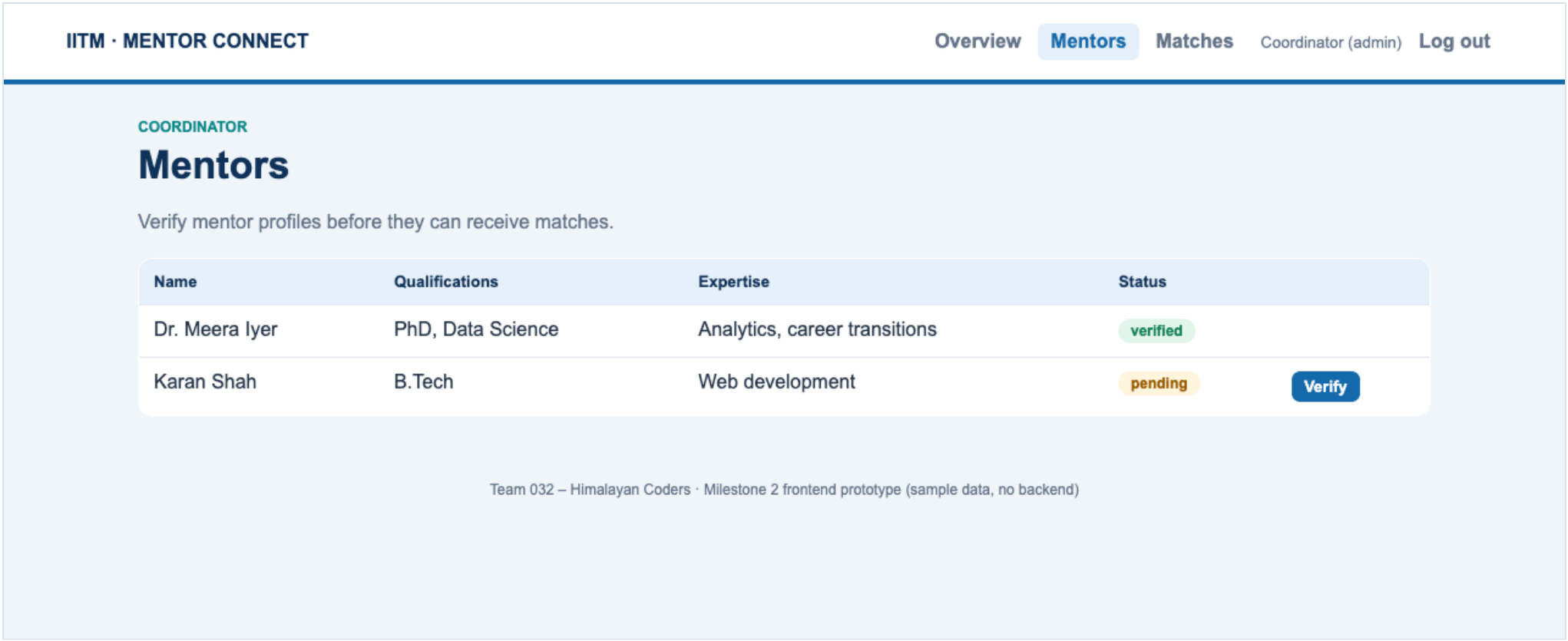


Figure 10. Mentor verification.

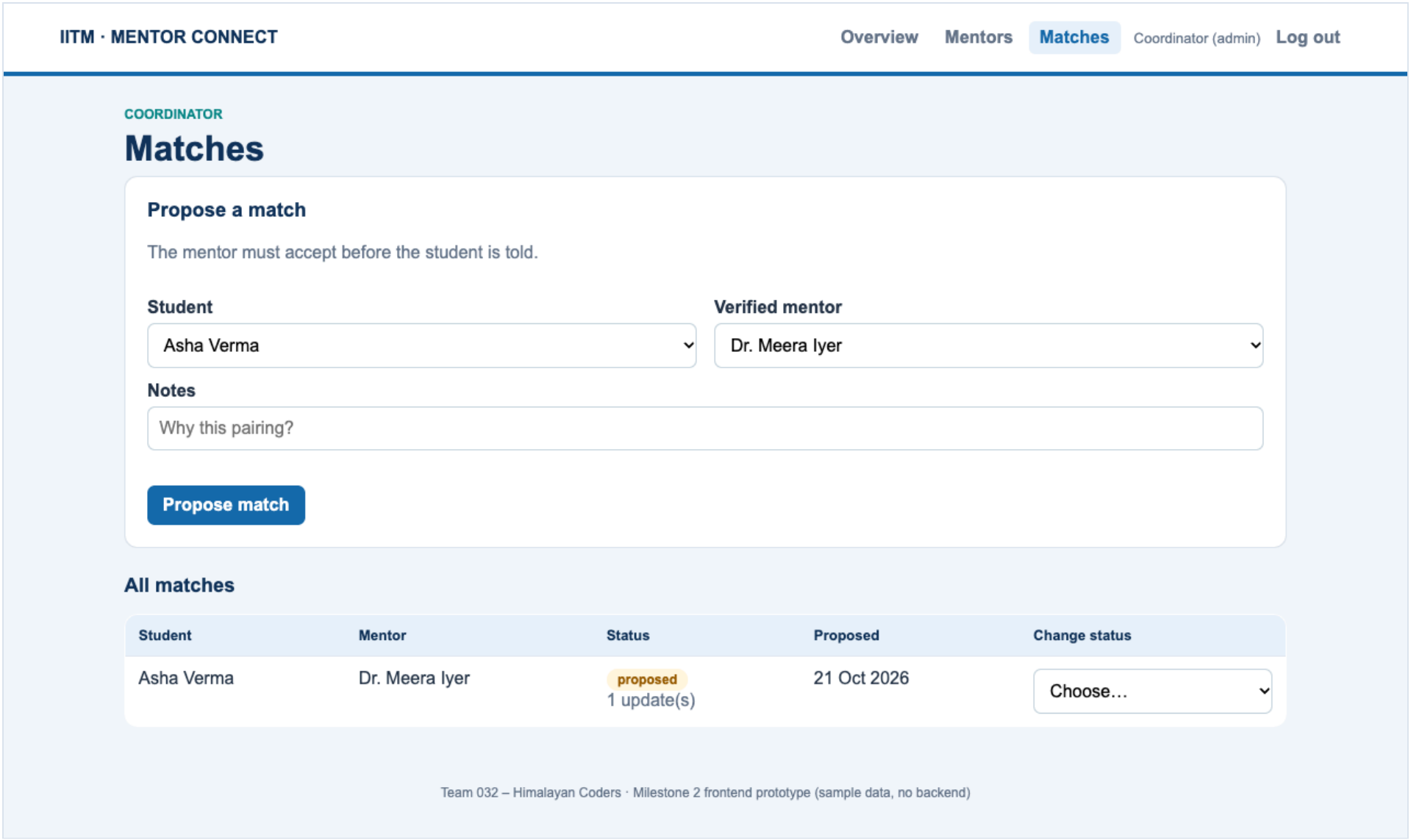


Figure 11. Proposing matches and changing their status.

7. Frontend Code and README

The frontend ZIP contains the 11 pages, `styles.css`, `app.js` and a README. To run it, unzip, open the `frontend` folder and open `index.html` in a modern browser. The demo password is `demo123` for the student, mentor and coordinator accounts listed on the login page.

8. Open Questions and Risks

- Whether a mentor may decline a match, and what happens to the student after a decline (the prototype returns the student to the coordinator).
- Which channels are used for notifications, and whether email is needed.
- How long student and mentor data is kept, and who may see feedback.
- Time zone handling for availability slots.